文献汇报

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Detect to Track and Track to Detect

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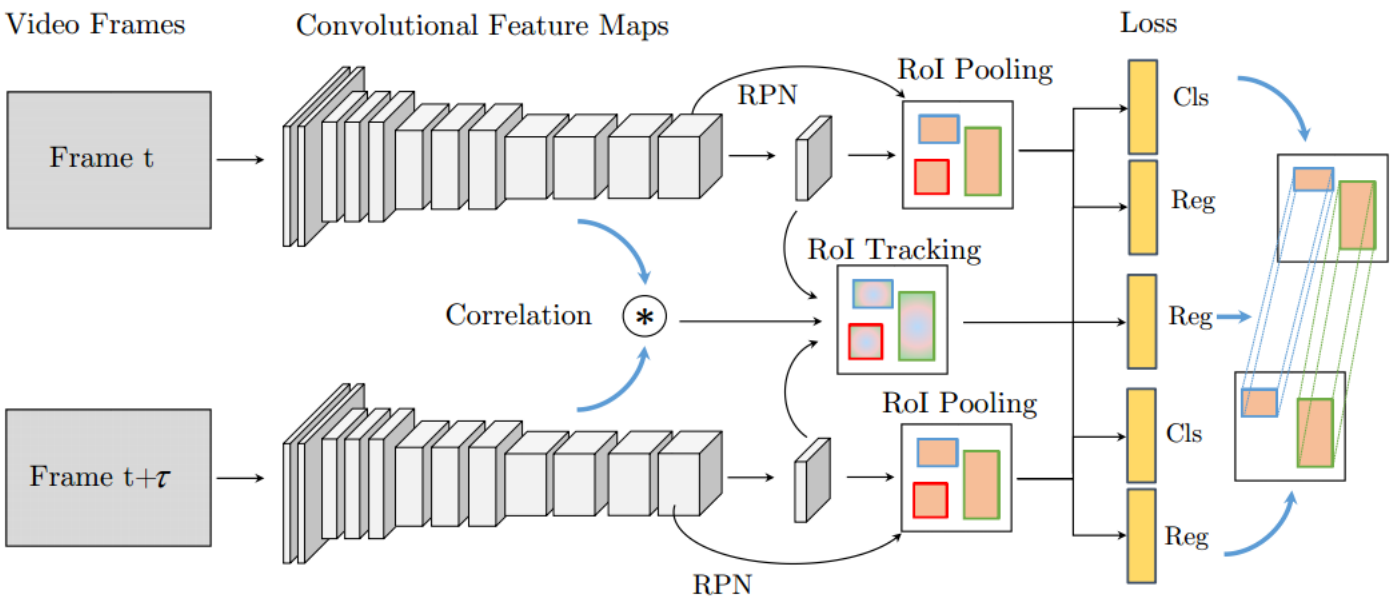


Figure 2. Architecture of our Detect and Track (D&T) approach (see Section 3 for details).

背景

视频目标检测最近掀起了一阵研究热潮，尤其是在ImageNet VID竞赛提出之后。不同于ImageNet DET竞赛，VID中目标会在图片序列中出现，并且往往伴随着额外的挑战：

1、数据集大小：VID中有1.3M张图片，作为对比，DET有400K张，COCO有100K张。
2、运动模糊：由相机运动或目标运动造成。
3、图片质量：网络视频片段的质量通常不如静态图片。
4、部分遮挡：由视角变化或者目标移动造成。
5、姿势：不常见的姿态在视频序列中频繁出现。

贡献

文章的贡献主要是三个方面：

1、设置了一个ConvNet结构来同时进行检测和跟踪，使用了一个多任务目标函数来进行逐帧的检测和跨帧跟踪回归。
2、引入了目标在时间尺度上的相关特征来辅助ConvNet进行跟踪。
3、我们将帧与帧之间的检测结果用跨帧的目标跟踪轨迹联系起来，以此来产生视频上的精确度更高的检测结果。

D&T方法

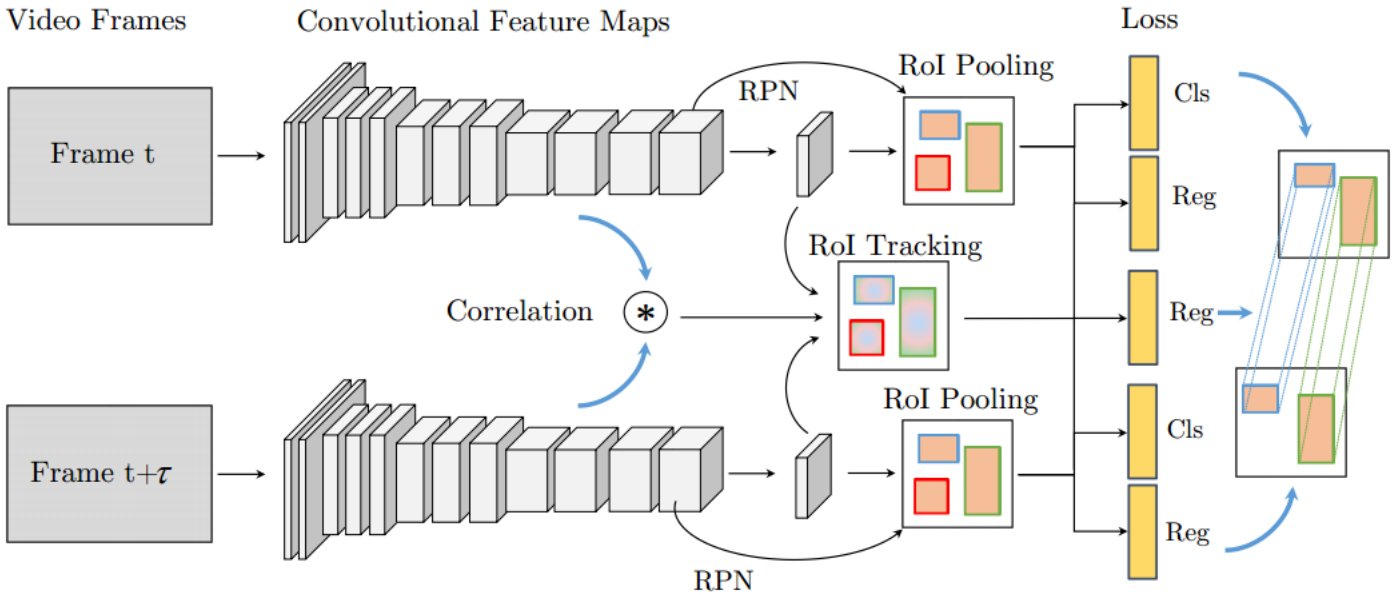
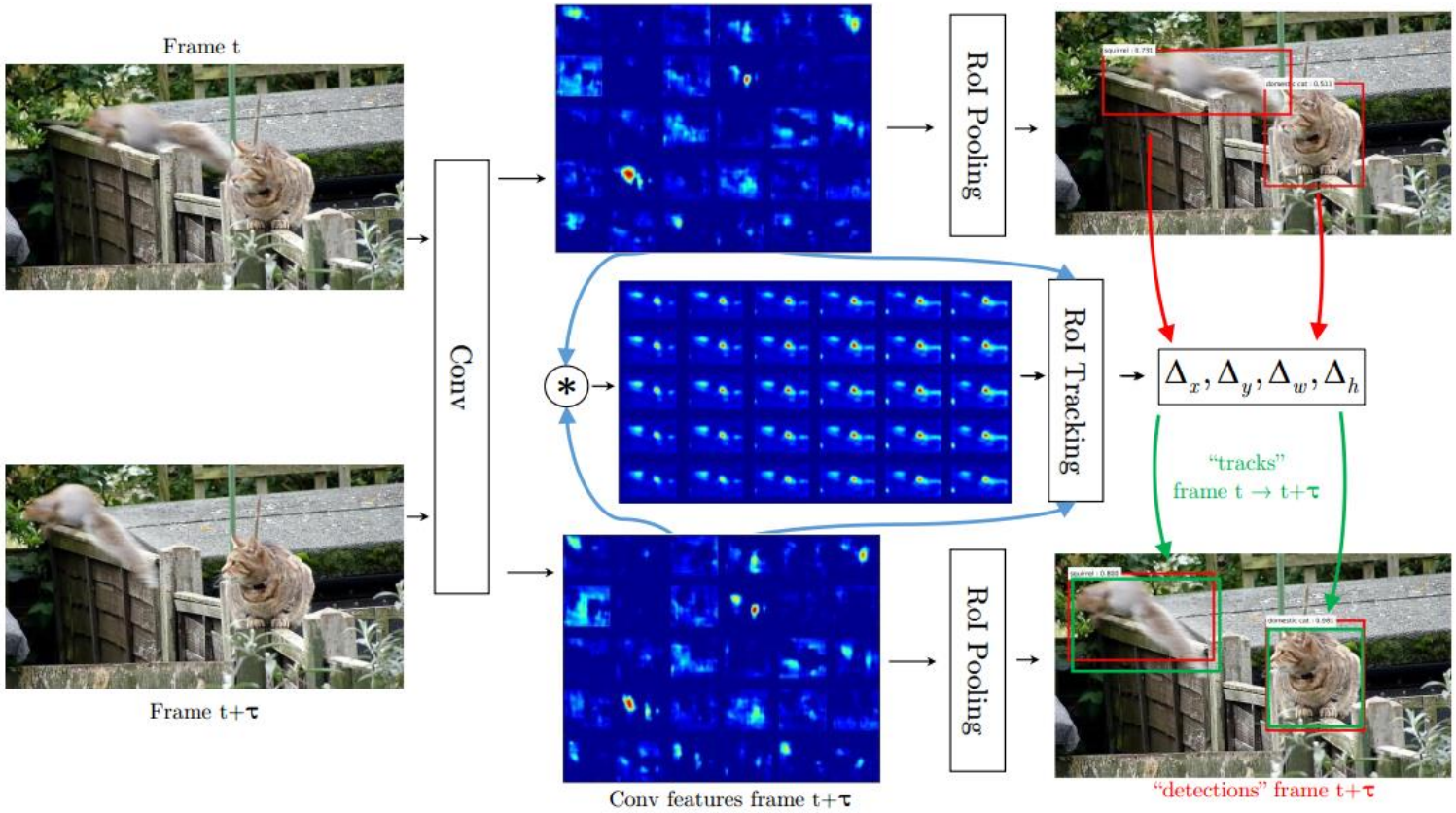


Figure 2. Architecture of our Detect and Track (D&T) approach (see Section 3 for details).

首先，给定一个图像对，通过一个特征提取网络（Res-101）提取深度特征供后序任务使用。一个RPN网络根据目标性大小输出候选区域，这些候选区域是对anchor的回归。ROI Pooling从中间的卷积层中生成位置敏感的特征图，用于最终的分类和回归。上述其实是R-FCN的框架，本文对上述框架进行了拓展，将两个PSROI Pooling得到的特征图，连同一个相关特征图，输入ROI tracking层，这个层输出的是框从前帧到后帧的变换。两帧之间的相关性，通过两帧之间PSROI Pooling之间计算得到。ROI tracking的训练，通过帧间的框的坐标回归得到。回归的损失是跟踪预测框与ground truth之间计算soft L1 norm得到。

D&T方法


Frame t
Frame t+τ
Conv
RoI Pooling
RoI Tracking
RoI Pooling
Conv features frame t+τ
$\Delta_x, \Delta_y, \Delta_w, \Delta_h$
"tracks"
frame t → t+τ
"detections" frame t+τ

损失函数

$$L(\{p_i\},\{b_i\},\{\Delta_i\}) = \frac{1}{N}\sum_{i=1}^{N} L_{cls}(p_{i,c^*})$$
$$+\lambda\frac{1}{N_{fg}}\sum_{i=1}^{N}[c_i^* > 0]L_{reg}(b_i, b_i^*)$$
$$+\lambda\frac{1}{N_{tra}}\sum_{i=1}^{N_{tra}} L_{tra}(\Delta_i^{t+\tau}, \Delta_i^{*,t+\tau}).$$

$$\Delta_x^{*,t+\tau} = \frac{B_x^{t+\tau} - B_x^t}{B_w^t} \qquad \Delta_y^{*,t+\tau} = \frac{B_y^{t+\tau} - B_y^t}{B_h^t}$$
$$\Delta_w^{*,t+\tau} = \log(\frac{B_w^{t+\tau}}{B_w^t}) \qquad \Delta_h^{*,t+\tau} = \log(\frac{B_h^{t+\tau}}{B_h^t})).$$

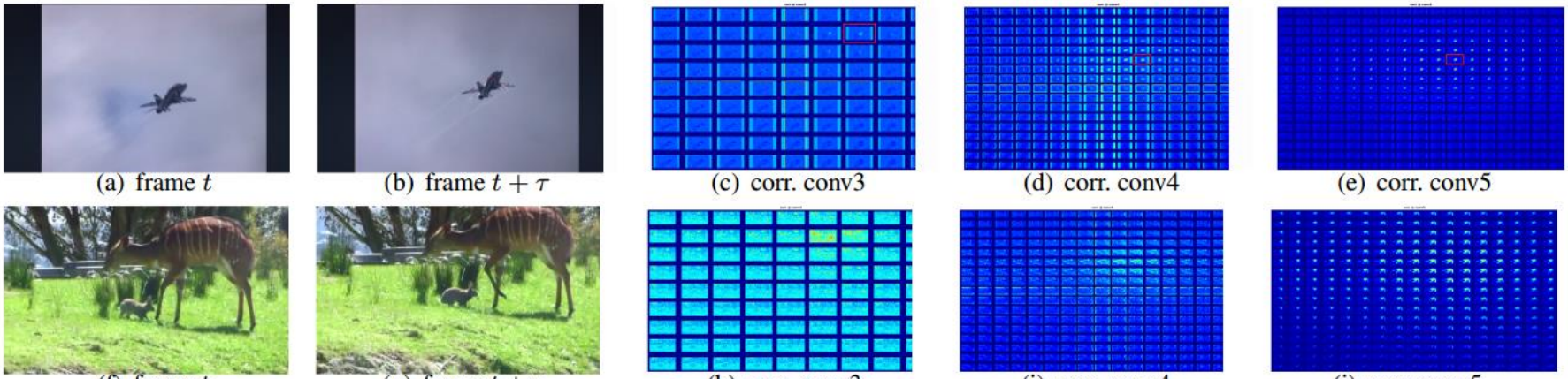
进行目标跟踪的相关特征

ments. Therefore, we restrict correlation to a local neighbourhood. This idea was originally used for optical flow estimation in [5], where a correlation layer is introduced to aid a ConvNet in matching feature points between frames. The correlation layer performs point-wise feature comparison of two feature maps $\mathbf{x}_l^t, \mathbf{x}_l^{t+\tau}$

$$\mathbf{x}_{corr}^{t,t+\tau}(i,j,p,q) = \left\langle \mathbf{x}_l^t(i,j), \mathbf{x}_l^{t+\tau}(i+p, j+q) \right\rangle \quad (4)$$

where $-d \leq p \leq d$ and $-d \leq q \leq d$ are offsets to compare features in a square neighbourhood around the locations i, j in the feature map, defined by the maximum displacement, d. Thus the output of the correlation layer is a feature map of size $\mathbf{x}_{corr} \in \mathbb{R}^{H_l \times W_l \times (2d+1) \times (2d+1)}$. Equation (4) can be seen as a correlation of two feature maps within a local square window defined by d. We compute this local correlation for features at layers conv3, conv4 and conv5 (we use a stride of 2 in i, j to have the same size in the conv3 correlation). We show an illustration of these features for two sample sequences in Fig. 4.

To use these features for track-regression, we let RoI pooling operate on these maps by stacking them with the bounding box features in Sect. 3.2 $\{\mathbf{x}_{corr}^{t,t+\tau}, \mathbf{x}_{reg}^t, \mathbf{x}_{reg}^{t+\tau}\}$.



(a) frame t (b) frame $t+\tau$ (c) corr. conv3 (d) corr. conv4 (e) corr. conv5

(f) frame t (g) frame $t+\tau$ (h) corr. conv3 (i) corr. conv4 (j) corr. conv5

Figure 4. Correlation features for two frames of two validation videos. For the frames in (a) & (b) , we show in (c),(d) and (e) the correlation maps computed by using features from conv3, conv4 and conv5, respectively. The feature maps are shown as arrays with the centre map corresponding to zero offsets p, q between the frames and the neighbouring rows and columns correspond to shifted correlation maps of increasing p, q. We observe that the airplane moves to the top-right; hence the feature maps corresponding to $p = 2, q = 3$ show strong responses (highlighted in red). Note that the features at conv4 and conv5 have the same resolution, whereas at conv3 we use stride 2 correlation sampling to produce equal sized outputs. In (h),(i) and (j) we show additional multiscale correlation maps for the frames in (f) & (g) which are affected by camera motion resulting in correlation patterns that correctly estimate this at the lower layer (conv3 corr. responds on the grass and legs of the animal (h)), and also handles the independent motion of the animals at the higher conv5 corr (j).

Linking tracklets to object tubes

Consider the class detections for a frame at time t, $D_i^{t,c} = \{x_i^t, y_i^t, w_i^t, h_i^t, p_{i,c}^t\}$, where $D_i^{t,c}$ is a box indexed by i, centred at (x_i^t, y_i^t) with width w_i^t and height h_i^t, and $p_{i,c}^t$ is the softmax probability for class c. Similarly, we also have tracks $T_i^{t,t+\tau} = \{x_i^t, y_i^t, w_i^t, h_i^t; x_i^t + \Delta_x^{t+\tau}, y_i^t + \Delta_y^{t+\tau}, w_i^t + \Delta_w^{t+\tau}, h_i^t + \Delta_h^{t+\tau}\}$ that describe the transformation of the boxes from frame t to $t+\tau$. We can now define a class-wise linking score that combines detections and tracks across time

$$s_c(D_{i,c}^t, D_{j,c}^{t+\tau}, T^{t,t+\tau}) = p_{i,c}^t + p_{j,c}^{t+\tau} + \psi(D_i^t, D_j, T^{t,t+\tau}) \tag{5}$$

where the pairwise score is

$$\psi(D_{i,c}^t, D_{j,c}^{t+\tau}, T^{t,t+\tau}) = \begin{cases} 1, & \text{if } D_i^t, D_j^{t+\tau} \in T^{t,t+\tau}, \\ 0, & \text{otherwise.} \end{cases} \tag{6}$$

Here, the pairwise term ψ evaluates to 1 if the IoU over-

Linking tracklets to object tubes

lap a track correspondences $T^{t,t+\tau}$ with the detection boxes $D_i^t, D_j^{t+\tau}$ is larger than 0.5. This is necessary, because the output of the track regressor does not have to exactly match the output of the box regressor.

The optimal path across a video can then be found by maximizing the scores over the duration $\mathcal{T}$ of the video [11]

$$\bar{D}_c^\star = \underset{\bar{D}}{\operatorname{argmax}} \frac{1}{\mathcal{T}} \sum_{t=1}^{\mathcal{T}-\tau} s_c(D^t, D^{t+\tau}, T^{t,t+\tau}). \qquad (7)$$

Eq. (7) can be solved efficiently by applying the Viterbi algorithm [11]. Once the optimal tube $\bar{D}_c^\star$ is found, the detections corresponding to that tube are removed from the set of regions and (7) is applied again to the remaining regions.

Methods	airplane	antelope	bear	bicycle	bird	bus	car	cattle	dog	d. cat	elephant	fox	g. panda	hamster	horse	lion
TCN [18]	72.7	75.5	42.2	39.5	25.0	64.1	36.3	51.1	24.4	48.6	65.6	73.9	61.7	82.4	30.8	34.4
TPN+LSTM [16]	84.6	78.1	72.0	67.2	68.0	80.1	54.7	61.2	61.6	78.9	71.6	83.2	78.1	91.5	66.8	21.6
Winner ILSVRC'15 [17]	83.7	85.7	84.4	74.5	73.8	75.7	57.1	58.7	72.3	69.2	80.2	83.4	80.5	93.1	84.2	67.8
D (R-FCN)	87.4	79.4	84.5	67.0	72.1	84.6	54.6	72.9	70.9	77.3	76.7	89.7	77.6	88.5	74.8	57.9
D (& T loss)	89.4	80.4	83.8	70.0	71.8	82.6	56.8	71.0	71.8	76.6	79.3	89.9	83.3	91.9	76.8	57.3
D&T ($\tau = 10$)	89.1	79.8	87.5	68.8	72.9	86.1	55.7	78.6	76.4	83.4	82.9	97.0	85.0	96.0	82.2	66.0
D&T ($\tau = 1$)	90.2	82.3	87.9	70.1	73.2	87.7	57.0	80.6	77.3	82.6	83.0	97.8	85.8	96.6	82.1	66.7
D&T (Inception-v4)	91.5	81.4	91.4	76.4	73.9	86.8	57.8	81.8	83.0	92.0	81.3	96.6	84.3	98.0	83.8	78.1

Methods	lizard	monkey	motorcycle	rabbit	red panda	sheep	snake	squirrel	tiger	train	turtle	watercraft	whale	zebra	mAP (%)
TCN [18]	54.2	1.6	61.0	36.6	19.7	55.0	38.9	2.6	42.8	54.6	66.1	69.2	26.5	68.6	47.5
TPN+LSTM [16]	74.4	36.6	76.3	51.4	70.6	64.2	61.2	42.3	84.8	78.1	77.2	61.5	66.9	88.5	68.4
Winner ILSVRC'15 [17]	80.3	54.8	80.6	63.7	85.7	60.5	72.9	52.7	89.7	81.3	73.7	69.5	33.5	90.2	73.8
Winner ILSVRC'16 [41]	(single model performance)														76.2
D (R-FCN)	76.8	50.1	80.2	61.3	79.5	51.9	69.0	57.4	90.2	83.3	81.4	68.7	68.4	90.9	74.2
D (& T loss)	79.0	54.1	80.3	65.3	85.3	56.9	74.1	59.9	91.3	84.9	81.9	68.3	68.9	90.9	75.8
D&T ($\tau = 10$)	83.1	57.9	79.8	72.7	90.0	59.4	75.6	65.4	90.5	85.6	83.3	68.3	66.5	93.2	78.6
D&T ($\tau = 1$)	83.4	57.6	86.7	74.2	91.6	59.7	76.4	68.4	92.6	86.1	84.3	69.7	66.3	95.2	79.8
D&T (Inception-v4)	87.9	59.4	89.3	78.6	94.7	63.3	80.1	71.9	92.5	84.6	85.5	70.1	67.4	95.2	**82.0**

Table 1. Performance comparison on the ImageNet VID validation set. The average precision (in %) for each class and the mean average precision over all classes is shown. τ corresponds to the temporal sampling stride. Our D&T variants use ResNet-101 [12] as backbone, except for the last row which lists performance for an Inception-v4 [37] backbone which excels at some challenging object categories.

Backbone	D	D&T	D&T, average
ResNet-50	72.1	76.5	76.7
ResNet-101	74.1	79.8	80.0
ResNeXt-101-32×4	75.9	81.4	81.6
Inception-v4	77.9	82.0	**82.1**

Table 2. Backbone network comparison for image-based **D**etection, and video-based **D**etection & **T**racking architectures. mAP (in %) over all classes on ImageNet VID validation is shown.